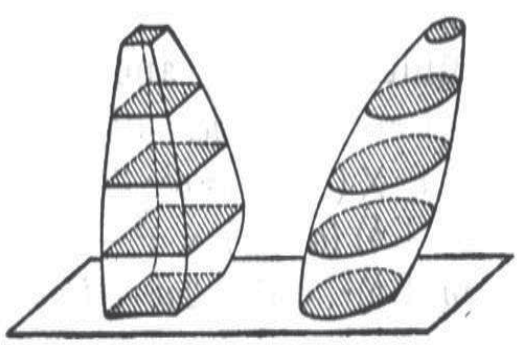
Ratio - a relationship between two things that have been compared; normally expressed as the quotient of one divided by the other. The ratio of 4 to 7 is written 4 : 7 or 4 to 7 or 4/7.

You might say 4/7 looks like a fraction. That is because it really is a fraction. Example: if you have 25 problems on a math test and you missed 2, the ratio of incorrect problems to the total number of problems would be 2 to 25 (which can be written 2:25 and 2/25). If the ratio was written the other way around (25:2) the ratio would mean 25 problems, you missed 2. The number that is listed first is explained first.

Definitions

Trace the dotted numbers and words.

Percent - one part in a hundred. when you think of percent think "per hundred." Example: if your ratio is 2:25, the percent that you missed would be 8%. this means you missed 8 out of 100 problems. If you had a test with 100 problems, you would have missed 8 of the problems.



Review-

A ratio is a relationship of two things when they are compared.

A percent means "per hundred."

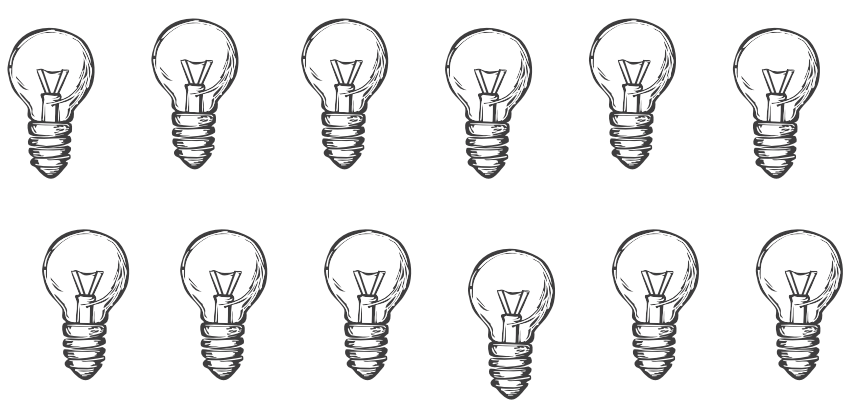
Definitions

Trace the dotted symbols and words.

1. Pat spent four of her ten coins to buy a lollipop.
Circle the ratio 4:10 coins.

2. Mary picked five out of the eight tomatoes in the basket.
Circle the ratio 5:8 tomatoes.

3. Fred removed six out of twelve light bulbs from their package.
Circle the ratio 6:12 light bulbs.



1. Circle 4 coins 2. Circle 5 tomatoes 3. Circle 6 bulbs

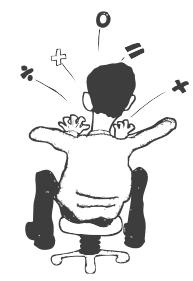
Let's Practice

Cirlce the ratio amount.

1. A ratio is a_____ between two things that have been compared.

2. A ratio is normally expressed as the _____of one divided by the other.

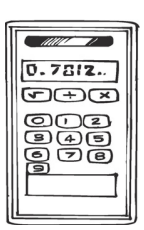
3. When writing a ratio a number that is listed first is _____ first.



4. When you think of percent, think per_____.

5. The ratio of number of problems missed (2) to the total number of problems (10) can be written_____.

6. A ratio can look like a _____ because it really is one.



Ratios

Please answer the questions.

1. A ratio is a_____ between two things that have been compared. **relationship**

2. A ratio is normally expressed as the _____of one divided by the other. **quotient**

3. When writing a ratio a number that is listed first is _____ first.

explained

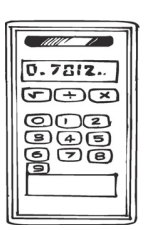
4. When you think of percent, think per_____.

hundred

5. The ratio of number of problems missed (2) to the total number of problems (10) can be written_____.

2:10

6. A ratio can look like a _____ because it really is one. **fraction**



d	d	e	s	n	r	r	c	u
e	e	s	d	e	p	n	o	t
r	r	s	d	p	o	e	n	a
d	a	o	s	t	c	e	p	p
n	p	e	p	e	i	e	h	a
u	m	t	r	t	r	c	h	r
h	o	e	o	c	a	p	d	p
r	c	u	e	r	t	r	x	d
e	q	n	e	n	i	e	u	e
p	t	p	r	o	o	t	c	m

Word Bank

ratio
percent
compared
expressed
quotient
per hundred

Word Search

Find and circle the words in the Word Bank.

It's not difficult to write ratios. we do have to listen to how things are listed. In the explanation whatever is explained first is written first in your ratio. If we say "3 balls out of 7 balls" we would write that ratio as 3:7 or 3/7 or 3 to 7.

Remember: "whatever is explained first is written first". When the ratio is listed as "out of 7 balls", 3 balls were white, we would write that ratio as 7:3 or 7/3 or 7 to 3. Remember: "whatever is explained first is written first".

Writing Ratios

Read the information and trace the dotted words and numbers.

It is very important to listen to how the ratio is listed. Remember what is explained first is written first. In a ratio, we don't even have to compare the same things. If we are talking about what was served for breakfast, we could compare eggs to pancakes. If the sixth grade class had breakfast and 412 students chose pancakes and 225 students chose eggs to eat, we could still write that as a ratio. Listen to how things are listed. What is the ratio of eggs to pancakes -225 eggs and 412 pancakes, so our ratio of eggs to pancakes would be 225:412. But if we wanted to know the ratio of pancakes to eggs, (listen to how it is listed) we would write that ratio as 412:225

Writing Ratios

Read the information and trace the dotted words and numbers.

Problem: The sixth grade class had a field day at their school. Two-hundred-thirty-four students had their faces painted with the American Flag and three-hundred-twenty-two students had their faces painted with their school's mascot.

Question: What is the ratio of children with the American Flag to the children with their school's mascot?

American Flag = 234 students

School's mascot= 322 students

Flag to mascot

Flag to mascot= 234:322

Question: What is the ratio of children with their school's mascot to the children with the American Flag?

American Flag= 234 students

School's mascot= 322 students

Mascot to Flag

Mascot to Flag = 322:234

1. It is not _____to write_____.

2. We _____have to _____how things are listed.

3. Remember, "Whatever is _____first is _____first".

4. If we list the number of basketballs first, then the number of basketballs will be _____first in our ratio.

5. It is _____important to listen to how the ratio is listed.

6. In a ratio, we _____even have to compare the same things.

7. Sixty-nine bicycles and seventy skateboards -What is the ratio of skateboards to bicycles?_____

8. Forty-four cookies and fifty-five cupcakes -What is the ratio of cupcakes to cookies?_____

9. Sixteen chairs and twenty-two desks -What is the ratio of desks to chairs?_____

1. It is not _____ to write _____.

difficult **ratios**

2. We _____ have to _____ how things are listed.

do **listen**

3. Remember, "Whatever is _____ first is _____ first".

explained **listed**

4. If we list the number of basketballs first, then the number of basketballs will be _____ first in our ratio.

written

5. It is _____ important to listen to how the ratio is listed.

very

6. In a ratio, we _____ even have to compare the same things.

don't

7. Sixty-nine bicycles and seventy skateboards -What is the ratio of skateboards to bicycles? _____

70:69

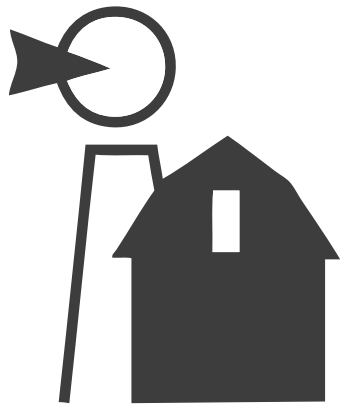
8. Forty-four cookies and fifty-five cupcakes -What is the ratio of cupcakes to cookies? _____

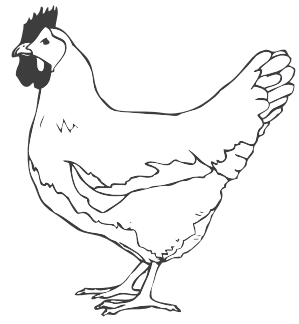
55:44

9. Sixteen chairs and twenty-two desks -What is the ratio of desks to chairs? _____

22:16

Put the animals in the correct barn by writing the ratio.





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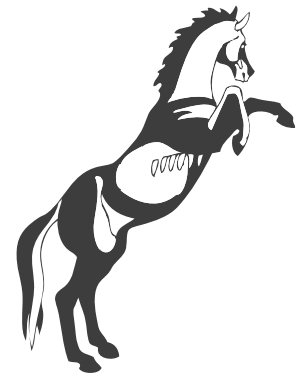


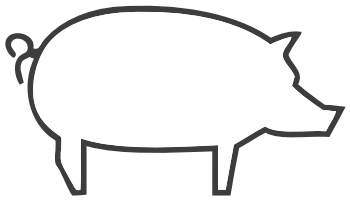
12

A. Cows to sheep

_____

B. Turkeys to Chickens

_____



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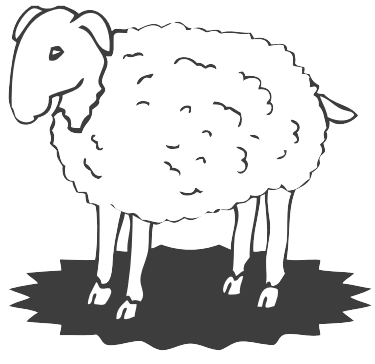

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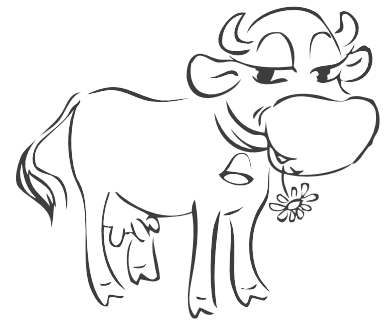

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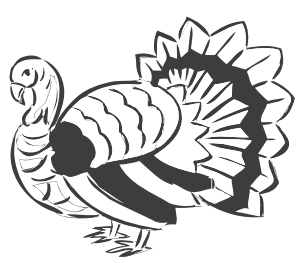

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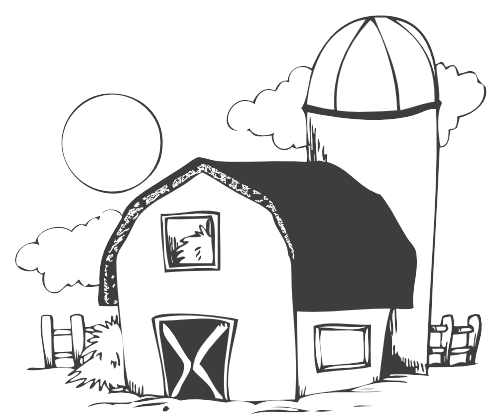


C. Horses to pigs

_____

D. Pigs to Chickens

_____

Barn A	14:8	Barn C	12:9
Barn B	7:15	Barn D	9:15

It is sometimes easier to understand ratios if we can see some concrete models. We have seven balls and we want to compare how many footballs are in a group of balls. Look at the pictures below.

You see seven balls. Remember that there are seven balls in all. How many balls are footballs? There are three footballs in the group. So, the ratio of football to the total number of balls is 3 : 7. Remember we have learned the first number explained is the first number written. It is very important to read how the number should be written. Remember :

First explained
first written.

Concrete Models

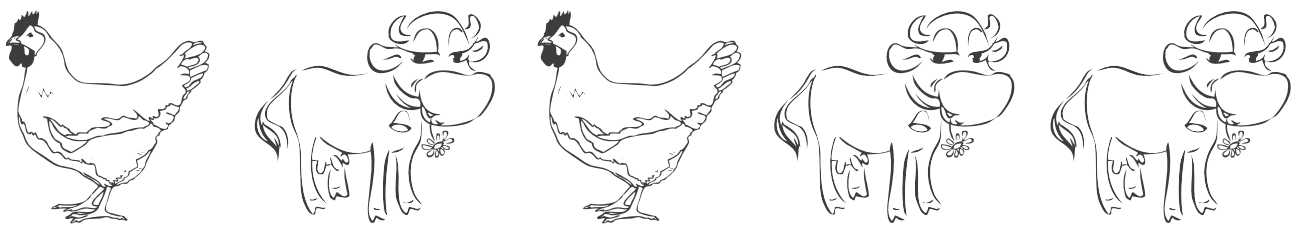
Read the information and trace the dotted words and numbers .

1.

How many total flowers are there in our group? 8

How many flowers are roses? 3

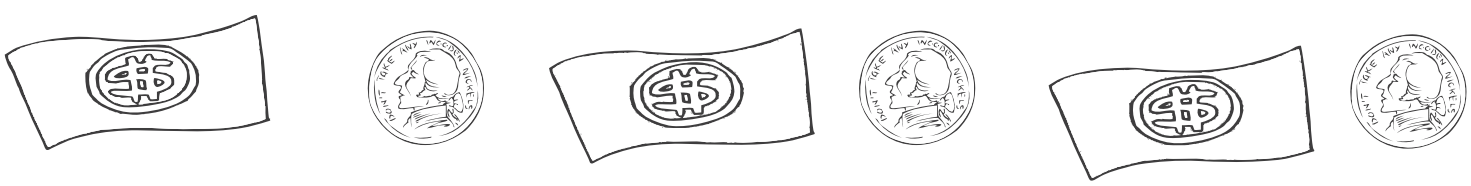
To write the ratio of how many roses are in the total group, you could write 3:8.

2 

How many total animals are there in our group? 5

How many animals are chickens? 2

To write a ratio of how many chickens are in the total group, you could write 2:5.

3. 

How many total coins and bills are there in our group? 6

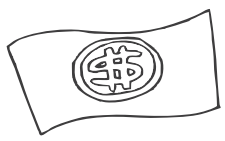
How many coins are there in the group? 3

To write a ratio of how many coins are in the total group, you could write 3:6.

1. Sometimes it is easier to understand ratios if we can see some

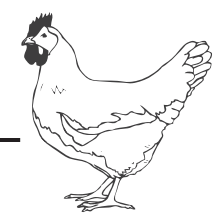
_____models.

2. A ratio is when things are _____.



3. When we write a comparison of the numbers of footballs to

the total number of balls we write it in a _____



4. Remember we have learned the _____number

explained is the _____number written.

5. A comparison written like 5:9 is known as a _____.

6. In a ratio we need to know the total things in

a_____ and how many we are talking about.

Ratio Quiz

Answer the questions.

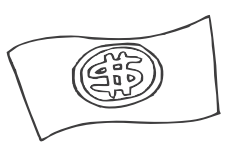
1. Sometimes it is easier to understand ratios if we can see some

_____models.

concrete

2. A ratio is when things are _____.

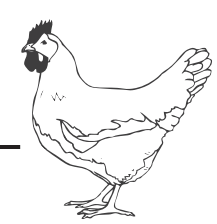
compared



3. When we write a comparison of the numbers of footballs to

the total number of balls we write it in a _____

rati



4. Remember we have learned the _____number

first

explained is the _____number written.

first

5. A comparison written like 5:9 is known as a _____.

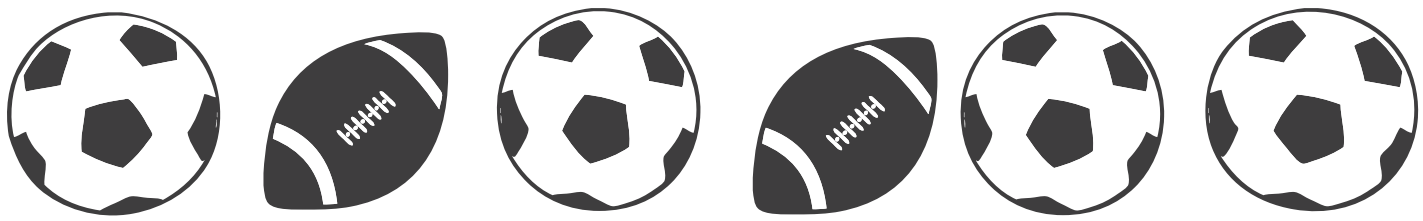
ratio

6. In a ratio we need to know the total things in

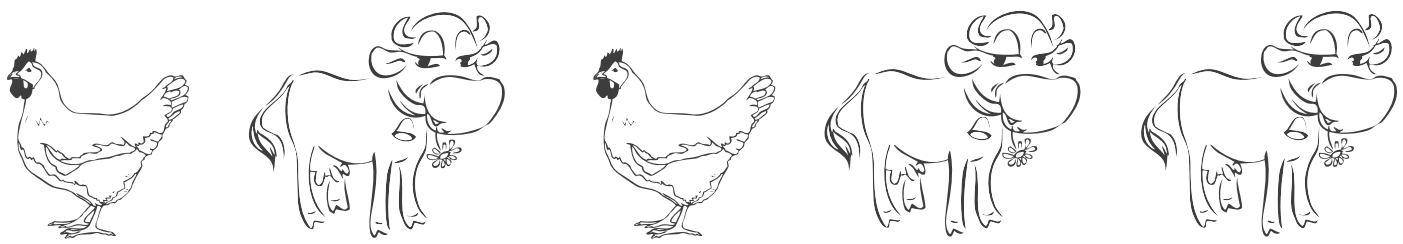
a_____ and how many we are talking about.

group

A.



B.



C.

1. Which letter shows the ratio 2: 5? _____

2. Which letter shows the ratio 4: 7? _____

3. Which letter shows the ratio 4: 6? _____

1. B 2. C 3. A

Ratios

Determine the ratios.

You can also represent ratios as fractions. You can think of any fraction as a ratio. Remember that a fraction is "part" over "whole". The ratio compares "part" to the "whole". So, the ratio 6:10, can be written as the fraction 6/10.

Problem:

Joey hit 11 out of 20 balls that were thrown to him in the baseball game. What fraction represents his hits to balls thrown?

Eleven is the "part" and twenty is the "whole".

The ratio is 11:20 and the fraction is 11/20.

Problem:

Meera made 6 of the 10 shots she took at hockey practice yesterday. What fraction represents her shots made to total shots? Six is the "part" and ten is the "whole". The ratio is 6:10 and the fraction is 6/10

1. Evan played a computer game and killed 19 of the 25 enemies. What was his ratio (written as a fraction) of enemies killed to total enemies?

2. Eric took his reading test today. He missed 2 questions out of 50 questions on the test. What was his ratio (written as a fraction) of questions missed to total questions?

3. Charlie was reading his library book for homework. He read 50 pages when his homework assignment was 75 pages. What was his ratio (written as a fraction) of pages read to the total number of pages he was supposed to read?

4. Beverly was buying apples to bake pies. She chose 10 red apples out of the 25 apples she bought. What was her ratio (written as a fraction) of red apples to the total number of apples?

5. Susan went sailing 4 days this week. What was her ratio (written as a fraction) of the days she went sailing to the total number of days in a week?

1. 19/25 2. 2/50 3. 50/75 4. 10/25 5. 4/7

1. You can represent ratios as a _____.

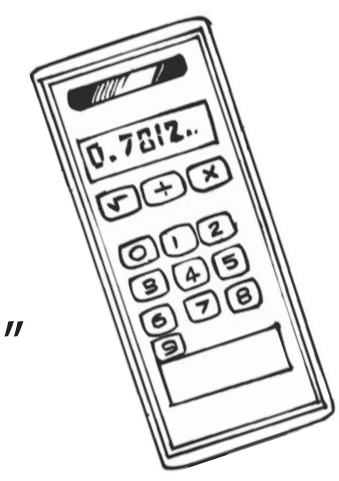
2. You can think of any fraction as a _____.

3. Remember that a fraction is " _____ " over " whole. "

4. A ratio _____"part" to " whole ."

5. Remember: "whatever is _____

first is _____ first."



1. You can represent ratios as a _____.

fraction

2. You can think of any fraction as a _____.

ratio

3. Remember that a fraction is " _____ " over " whole."

part

4. A ratio _____"part" to " whole ."

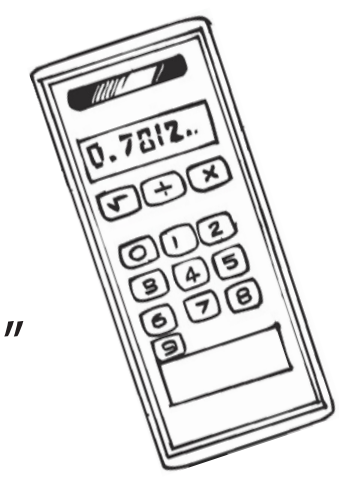
compares

5. Remember: "whatever is _____

explained

first is _____ first."

written



The underlined word has been scrambled. Please rearrange the letters. Write the correct word on the line to the right.

1. A ratio mepacors "part" to "whole." _____

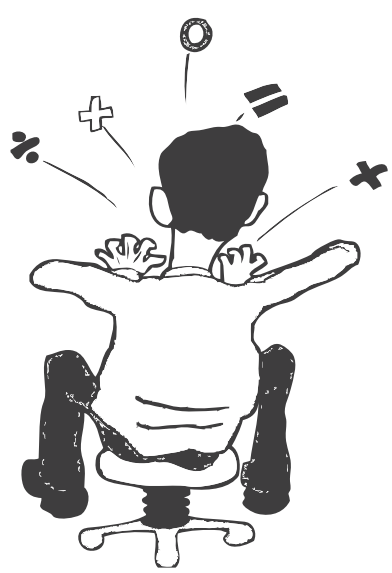
2. You can represent ratios as trafisnoc. _____

3. The toiar 4:9 can be written as the fraction 4/9. _____

4. A fraction is "part" over "helow." _____

5. A ratio and a rionacft look alike. _____

6. A trafcion and a taroi both compare "part" to "whole." _____



_____

We can change ratios into percents whenever it is needed. If we put the ratio into a fraction, we can change the fraction into a percent really easily. We can take the fraction 6/10 and convert it into the percent of 60% by changing 6/10 into ?/100. Remember that when we talk about "percent" we should think "per hundred." If the fraction turns out to be so many out of ten, then multiply each number in the fraction by ten to make the denominator one hundred. When we have a number over 100 we would have our percent. If we have a ratio of 15:20, change this ratio into the fraction 15/20.Ask yourself what you have to multiply 20 by to make it 100. The answer would be 5(5X20=100). When you multiply the 15 by 5(15X5) also, you get 75. So our percent is 75%. Don't forget when you think of "percent" think "per hundred" We always want our denominator to be 100 so it will be easy to see what our percentage would be.

Percents

Read the information, then trace the dotted words and numbers.

Remember that we want our fractions to have denominators to be 100.

1. Our ratio is 18:25 - Our fraction would be 18/25. We would multiply 25X4 to make 100. 18X4=72; so our fraction would be 72/100 and our percent would be 72%.

2. Our ratio is 43:50 - Our fraction would be 43/50. We would multiply 50X2 to make 100. 43X2=86. So our fraction would be 86/100 and our percent would be 86 %.

3. Our ratio is 8:10 - Our fraction would be 8/10. We would multiply 10X 10 to make 100. 8X10=80, so our fraction would be 80/100 and our percent is 80%.

4. Our ratio is 18:20 - Our fraction would be 18/20. We would multiply 20X 5 to make 100. 18X5=90, so our fraction would be 90/100 and our percent would be 90%.

5. Our ratio is 4: 5- Our fraction would be 4/5. We would multiply 5X 20 to make 100. 4X20=80, so our fraction would be 80/100 and our percent is 80%.

1. We can change ratios into _____.

2. We can change the _____ into percent.

3. Remember that when we talk about " percent " we should think " _____ ."

4. Change the ratio to a _____to find the percent.

5. If the fraction turns out to be 7/10, we would multiply the 10 by _____ to make 100.

6. If the fraction turns out to be 23/25, we would multiply the 25 by _____to make it 100.

7. To turn a fraction into a percent we would multiply the _____by whatever it takes to make 100.

8. If the fraction turns out to be 18/20, we would multiply the 20 by _____ to make it 100.

9. If the fraction turns out to be 4/5, we would multiply the 5 by _____to make it 100.

10. If the fraction turns out to be 48/50, we would multiply the 50 by _____to make 100.

1. We can change ratios into _____.

fractions

2. We can change the _____ into percent.

fraction

3. Remember that when we talk about " percent " we should think " _____ ."

per hundred

4. Change the ratio to a _____to find the percent.

fraction

5. If the fraction turns out to be 7/10, we would multiply the 10 by _____ to make 100.

10

6. If the fraction turns out to be 23/25, we would multiply the 25 by _____to make it 100.

4

7. To turn a fraction into a percent we would multiply the _____by whatever it takes to make 100.

denominator

8. If the fraction turns out to be 18/20, we would multiply the 20 by _____ to make it 100.

5

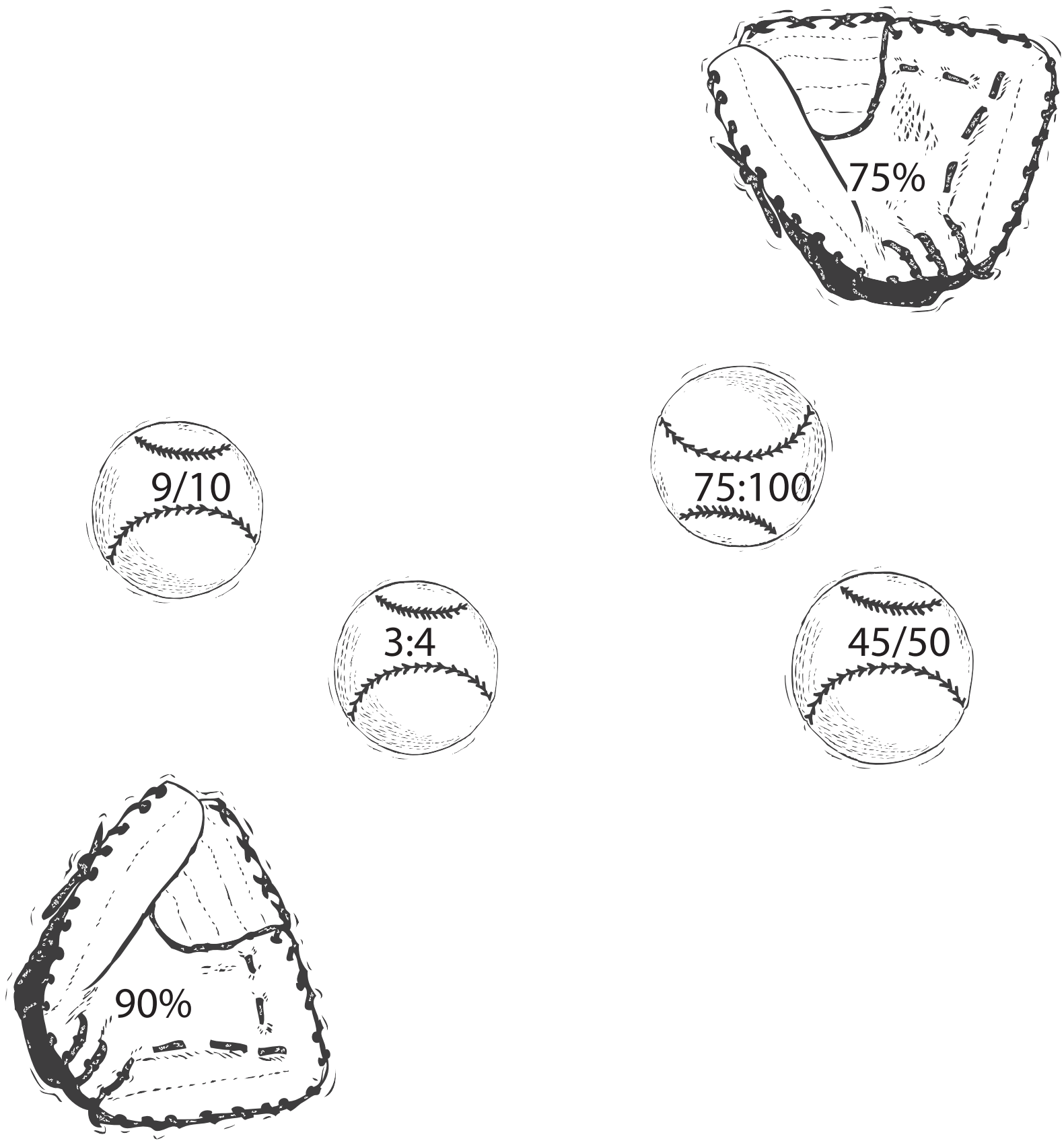
9. If the fraction turns out to be 4/5, we would multiply the 5 by _____to make it 100.

20

10. If the fraction turns out to be 48/50, we would multiply the 50 by _____to make 100.

2

Draw a line from the baseball to the correct baseball glove.



75%-3:4 75:100 90%-9/10 45/50

Converting ratios to decimals is a three step process.

Step one – We have learned that we can take a ratio that is expressed in the form using a colon (3 balls out of the total of 5 balls that are basketballs=3:5) and turn this expression into a fraction very easily.

Step two – After we have these ratios in a fraction, we then have to convert (change) these fractions into equivalent (equal) fractions with 10 as our denominator (the number on the bottom). So, if our denominator is 5, we would multiply both denominator and the numerator (number on top) by 2 to have a denominator of 10. If we use the ratio 3:5, we would convert (change) that into the fraction 3/5. Multiply both the numerator and the denominator by 2 to get the equivelent (equal) fraction 6/10.

Step three – When we say this fraction out loud we hear six tenths. Knowing our decimal places, we can write six tenths as .6 and this also reads as six tenths.

Converting ratios to decimals.

Read the information, then trace the dotted words and numbers.

If we have the expression 4:5,

(1) We would change that expression into the fraction 4/5.

(2) Then we could ask ourselves what we could multiply 5 by to make a ten? The answer is 2. Then we would multiply the denominator and numerator by 2 to have the fraction 8/10.

(3) That fraction would be read eight tenths. We can write eight tenths as the decimal .8.

If we have the expression 22:25,

(1) We would change that expression into the fraction 22/25.

(2) Then could we ask ourselves what we could multiply 25 by to make a hundred? The answer is 4.

Then we would multiply the denominator and numerator by 4 to have the fraction 88/100.

(3) That fraction would be read eighty-eight hundredths. We can easily write eighty-eight-hundredths as the decimal .88

1. Converting ratios to decimals is a _____ _____ process.

2. In Step One we take the ratio that is expressed in a form using a colon and turn it into a _____.

3. In Step Two we convert (change) that fraction into an _____ fraction with a denominator of 10 or 100.

4 In Step Three we say this fraction _____ _____ so we can hear it.

5. .After we hear the fraction out loud, we can write the _____.

6. 3:5=_____(the fraction)=_____(equivalent fraction)=_____(decimal).

7. 4:10 =_____(the fraction) =_____ (the decimal). We skipped Step_____.

8. 40/50 = _____ (the fraction) = _____(the decimal). We skipped Step_____.

Decimal Quiz *Answer the questions.*

1. Converting ratios to decimals is a _____ _____ process.
three **step**

2. In Step One we take the ratio that is expressed in a form using a colon and turn it into a _____.
fraction

3. In Step Two we convert (change) that fraction into an _____ fraction with a denominator of 10 or 100.
equivalent

4 In Step Three we say this fraction _____ _____ so we can hear it.
out **loud**

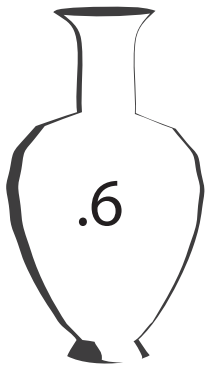
5. .After we hear the fraction out loud, we can write the _____.
decimal

6. 3:5=_____(the fraction)=_____(equivalent fraction)=_____(decimal).
3/5 **6/10** **.6**

7. 4:10 =_____(the fraction) =_____ (the decimal). We skipped Step_____.
4/10 **.4** **Two**

8. 40/50 = _____ (the fraction) = _____(the decimal). We skipped Step_____.
40/50 **80/100** **Two**

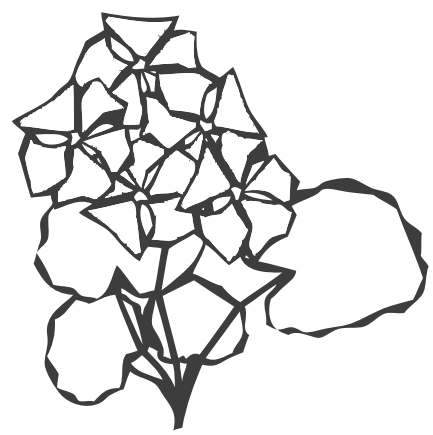
Draw a line from each flower to the vase it belongs in.



3:4

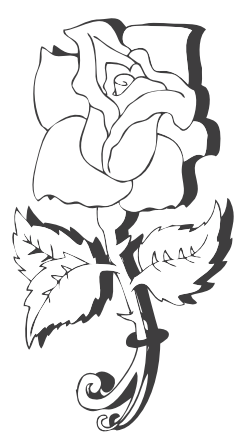
6/10

3:4



6:10

75/100



3/4

6 = 6:10, 6/10 .75 = 3:4, 3/4, 75/100

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